

Chao Wang^{PhD}

Human-robot Interaction Researcher | Human-centered robot learning

[Portfolio](#) | [LinkedIn](#) | [Google Scholar](#) | wallacewangchao@gmail.com

Research scientist and engineer specializing in multimodal human-robot interaction and human-centered robot learning. Builds real-time robotic systems that integrate language, vision, gaze, human action, and expressive behavior, and develops human demonstration and intervention pipelines for training, evaluating, and deploying safe, legible, and interactive robot behaviors in real-world environments.

EXPERIENCE

- Honda Research Institute, Principal Scientist / Engineer

SEP 2018 - PRESENT, Frankfurt, Germany (US EB-1A Approved · Priority Date Current)



■ **Multimodal Human-Robot Interaction:** Fine-tuned and integrated multimodal models to interpret **speech, gaze, visual context, and human actions**, enabling low-latency intent inference and coordinated **verbal** and **non-verbal** robot responses. Designed and implemented the AURA high-DoF expressive robot head to embody these responses through animated eyes, eyelids, ears, neck motion, gaze, and facial expression. [\[Project page\]](#)

■ **Contact-Rich Physical HRI & Tactile Interaction:** Led research integrating compliant robot **motion, torque sensing, tactile skin**, and visual and auditory feedback to support safe, legible, and expressive physical interaction between people and robots.

■ **XR Teleoperation & Robot-Learning Infrastructure:** Built an immersive teleoperation system via XR headset that retargets **human head, hand, gaze, body, and facial motion** to robot control while recording synchronized visual, robot-state, and action data for **imitation learning and VLA training**. Added close-proximity physical interaction support and operator-view compensation to improve stability, comfort, and demonstration quality. [\[Project page\]](#)

■ **Learning Expressive Robot Behavior from Demonstration in XR:** Developed an **XR demonstration pipeline** that captures synchronized human motion, gaze, facial expression, and interaction context. Trained flow-matching policies to generate real-time, context-aware robot expressions and non-verbal behaviors from human demonstrations. [\[Project page\]](#)

■ **XR APP for Reachy Mini Teleoperation:** Open-sourced a Unity/Meta Quest system for Reachy Mini teleoperation, streaming Quest head/body/hand tracking to robot motion commands and receiving live robot camera video via WebRTC. [\[Project page\]](#)

■ **Physical Robot Head & Face Design:** Designed and implemented the AURA robot head, a high-DOF expressive platform with animated **eyes, eyelids, ears, and neck**. Developed UI visualizations of robot attention and emotion, translating cognitive states into tangible cues.

■ **Human-Centered Data Collection for Robot Learning:** Designed and implemented a gamified XR platform that enables non-expert users to intuitively demonstrate household tasks in a virtual kitchen. The platform supports scalable collection of human demonstrations while preserving natural task strategies and behavior. [\[Project page\]](#)

■ **In-Vehicle AR Interfaces:** Designed AR-based driver-assistance UIs that blend spatial visualization with physical context. Deployed prototypes in simulators and test vehicles to evaluate driver comprehension and comfort.

■ **Human-Vehicle Collaboration Interfaces:** Connecting novel human-vehicle collaboration interfaces with algorithm for enhancing L3 autonomous driving, involving the integration of gaze, haptic feedback, and 3D visualization.

■ **Research Supervision & Academic Collaboration:** Led a Honda-funded PhD collaboration with LMU Munich and mentored PhD interns, master's students, and research interns in multimodal AI, VLA, social robotics, HRI, and interactive systems, from research proposal and prototyping to publication.

SKILLS

■ AI & Machine Learning:

Python, PyTorch, multimodal learning, VLM/LLM/VLA fine-tuning, flow matching, model evaluation, real-time inference

■ Human-AI Research:

HCI, HRI, human behavior modeling, experimental design, user studies, behavioral evaluation, mixed-method research, psychometrics

■ Robotics & Interactive Systems:

ROS/ROS2, Unity/C#, XR/VR, robot teleoperation, human motion and gaze tracking, WebRTC, multimodal sensing

■ Design:

Industrial Design, UI/UX, Rhino, Blender, Interaction Design, Figma

AWARDS & COVERAGES

■ **Best Paper Honorable Mention**, CHI 2025

■ **Best Paper Finalist**, ICRA 2024

■ **Best Demo Award**, AutomotiveUI 2017

■ **Best Poster Award**, AutomotiveUI 2016

■ **Best Paper Award**, AHFE 2016

■ **Red Dot Award**, NOMI, 2020

■ Coverages:

Featured by Engadget [\[Link\]](#) & New Scientist [\[Link\]](#)

PUBLICATIONS

■ 50+ peer-reviewed publications in CHI, IUI, IMWUT/UbiComp, AutomotiveUI, ICRA, IROS, HRI, and RO-MAN

PATENTS

■ 10+ filed patents, including 5 granted

- NIO, Creative Technologist

MAR 2017 - AUG 2018, Munich, Germany



■ **In-Car Robot “NOMI”:** Designed the expressive behavior of the world’s first production in-vehicle social robot, from emotional model to animated eye expressions. Conducted VR validation and deployment; shipped in 400,000+ vehicles. [\[Link\]](#)

■ **Autonomous Vehicle eHMI Design (EVE Concept):** Designed the external lighting interface and behavior for communication between AV and pedestrians; tested through immersive VR prototypes. [\[Link\]](#)

■ **Multimodal Driving Simulator:** Built an integrated simulator supporting haptic, gaze, speech, and visual interfaces, enabling rapid prototyping and evaluation of novel interaction concepts.

- Shanghai Moto Group, HMI Designer

MAR 2012 - AUG 2013, Shanghai, China



■ In-vehicle HMI systems design and corresponding interior component industrial design.

EDUCATION

Eindhoven University of Technology, PhD - HCI

SEP 2013 - NOV 2017, Eindhoven, Netherlands

Zhejiang University, M.A - Interaction/Industrial Design

SEP 2009 - MAR 2012, Hangzhou, China

SELECTED PUBLICATIONS

1. **Wang, C.**, Hasler, S., Tanneberg, D., Ocker, F., Joublin, F., Ceravola, A., ... & Gienger, M. (2024, May). **LaMI: Large Language Models for Multi-Modal Human-Robot Interaction.** In Extended Abstracts of the 2024 **CHI Conference** on Human Factors in Computing Systems (pp. 1-10). [\[website\]](#) [\[paper\]](#)
2. Leusmann, J., Belardinelli, A., Haliburton, L., Hasler, S., Schmidt, A., Mayer, S., ... & **Wang, C.** (2025, April). **Investigating LLM-Driven Curiosity in Human-Robot Interaction.** In Proceedings of the 2025 **CHI Conference** on Human Factors in Computing Systems (pp. 1-16). [\[website\]](#) [\[paper\]](#)
3. **Wang, C.**, Gienger, M., & Zhang, F. (2025). **Generation of Real-time Robotic Emotional Expressions Learning from Human Demonstration in Mixed Reality.** arXiv preprint arXiv:2508.08999. [\[website\]](#) [\[paper\]](#)
4. **Wang, C.**, Belardinelli, A., Tanneberg, D., Hasler, S., & Gienger, M. (2025). **Train your robot in AR: insights and challenges for humans and robots in continual teaching and learning.** *Frontiers in Robotics and AI*, 12, 1605652. [\[website\]](#) [\[paper\]](#)
5. Krüger, M., Tanneberg, D., **Wang, C.**, Hasler, S., & Gienger, M. (2025, August). **Mirror Eyes: Explainable Human-Robot Interaction at a Glance.** In 2025 34th IEEE International Conference on Robot and Human Interactive Communication (**RO-MAN**) (pp. 797-804). IEEE. [\[website\]](#) [\[paper\]](#)
6. Leusmann, J., Villa, S., Berberoglu, B., **Wang, C.**, & Mayer, S. (2025, April). **Developing and Validating the Perceived System Curiosity Scale (PSC): Measuring Users' Perceived Curiosity of Systems.** In Proceedings of the 2025 **CHI Conference** on Human Factors in Computing Systems (pp. 1-18). (Best paper honorable mention 🏆) [\[paper\]](#)
7. Joublin, Frank, Antonello Ceravola, Pavel Smirnov, Felix Ocker, Joerg Deigmoeller, Anna Belardinelli, **Chao Wang**, Stephan Hasler, Daniel Tanneberg, and Michael Gienger. **CoPAL: corrective planning of robot actions with large language models.** In 2024 IEEE International Conference on Robotics and Automation (**ICRA**), pp. 8664-8670. IEEE, 2024. (Best paper final list 🏆) [\[website\]](#) [\[paper\]](#)
8. Tanneberg, D., Ocker, F., Hasler, S., Deigmoeller, J., Belardinelli, A., **Wang, C.**, Wersing, H., Sendhoff, B., & Gienger, M. (2024). **To Help or Not to Help: LLM-based Attentive Support for Human-Robot Group Interactions.** 2024 IEEE/RSJ

International Conference on Intelligent Robots and Systems (**IROS**), 9130-9137. [[website](#) | [paper](#)]

9. **Wang, C.**, Belardinelli, A., Hasler, S., Stouraitis, T., Tanneberg, D., & Gienger, M. (2023, April). **Explainable human-robot training and cooperation with augmented reality**. In Extended Abstracts of the 2023 **CHI Conference** on Human Factors in Computing Systems (pp. 1-5). [[website](#) | [paper](#)]
10. Deigmoeller, J., Hasler, S., Agarwal, N., Tanneberg, D., Belardinelli, A., Ghoddoosian, R., **Chao Wang** ... & Gienger, M. (2025). **CARMA: Context-Aware Situational Grounding of Human-Robot Group Interactions by Combining Vision-Language Models with Object and Action Recognition**. In 2026 IEEE International Conference on Robotics and Automation (**ICRA**). [[paper](#)]
11. **Wang, C.**, Chu, D., Martens, M., Krüger, M., & Weisswange, T. H. (2022, September). **Hybrid Eyes: Design and Evaluation of the Prediction-Level Cooperative Driving with a Real-World Automated Driving System**. In Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (pp. 274-284). [[paper](#)]
12. Terken, J., Levy, P., **Wang, C.**, Karjanto, J., Yusof, N. M., Ros, F., & Zwaan, S. (2017). **Gesture-based and haptic interfaces for connected and autonomous driving**. In Advances in Human Factors and System Interactions: Proceedings of the AHFE 2016 International Conference on Human Factors and System Interactions, July 27-31, 2016, Walt Disney World®, Florida, USA (pp. 107-115). Springer International Publishing. (Best paper award 🏆) [[paper](#)]
13. **Wang, C.**, Terken, J., Hu, J., & Rauterberg, M. (2016, October). **"Likes" and "Dislikes" on the Road: A Social Feedback System for Improving Driving Behavior**. In Proceedings of the 8th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (pp. 43-50). [[website](#) | [paper](#)]